STAT 412

Module 9 Final Project

Note: Hyperlinks have been removed in this document. You may access these links on the Canvas page to navigate to the external sites or documents.

Activity Overview

On-time arrival data is an important consideration for any airline. On-time arrivals data is tracked by the U.S. Department of Transportation and affects airline profitability and customer satisfaction metrics.

For the final project, you will use the same dataset as that used for the midterm project; however, you will be selecting a new random sample of 250 rows. You will analyze on-time arrival data using confidence interval and hypothesis test analysis using Excel.

For a review of Excel, please refer to the Excel training videos in the Module 5 Midterm Project (Access the link on the Canvas page.). Also, you may refer to the Excel Tutorials page (Access the link on the Canvas page.).

Please read all instructions for this project very carefully, and if you have questions, contact your instructor immediately. It is required to use the data and specific instructions (provided on the following tabs) to complete the following Word document template.

- STAT 412 Final Project Template (DOCX) (Access the document on the Canvas page.)

Please carefully read the instructions. Then, you will submit the filled-in Word document template and also submit your Excel file with statistical calculations and graphs. Round all answers to one (1) decimal place unless otherwise specified.

BOTH FILES - Word and Excel - must be submitted to receive credit for this final project. Please review the rubric for detailed grading information.

Use the Student Lounge discussion if you need help with the technology and/or contact your instructor as quickly as possible if you encounter any technical difficulties.

Save your assignment using a naming convention that includes your first and last name and the activity number (or description). Do not add punctuation or special characters. Submit it by the posted due date. Review the rubric for detailed grading information.

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Data

Download the Excel file containing on-time arrival data for various airlines. The Bureau of Transportation Statistics (USDOT) (Access the link on the Canvas page.) publishes data on all manner of transportation-related records. This data was extracted from the Bureau of Transportation Statistics statistical data repository (USDOT) (Access the link on the Canvas page.) website.

In particular, you will select a sample of 250 rows of data from the Excel data file. For this Final Project, you will be focusing on Column M which is "ARR_DELAY." This column refers to the difference in minutes between scheduled and actual arrival times. Early arrivals show negative numbers.

SECTION 1: SELECT A SAMPLE

You will randomly select 250 rows of data for your analysis. **Note: This will be a new and different random sample compared to that used for the midterm project.**

You can use any method to randomly select 250 rows of data, such as the use of a random number table or the use of a random number generator (RANDOM.ORG) (Access the link on the Canvas page.). Here is one method for random sample selection using Excel:

a) In cell R2, enter the command =RAND(). This Excel command generates a random number between 0 and 1.
b) Drag or copy/paste the =RAND() command to all rows in the table. You should now have a random number in column R for all rows in the Table. Note that these random numbers will update to new random numbers anytime the Excel file is modified in any way. To avoid this, you can copy and paste Column R to column S using the "PASTE VALUES" command in Excel. This will result in Column S containing "frozen values" of random numbers.
c) Sort the dataset using Column S as the basis for the sort, in ascending order. You will keep the first 250 rows of data and delete all remaining rows in the table.

Instructions

Important Information

When calculating values using Excel, copy the entry from the function bar and enter it into the template that you will turn along with the Excel workbook.

SECTION 2: Confidence Interval for Mean (15 points total, 5 points for each part)

Imagine you are an engineer at the Department of Transportation, and you are tasked to derive an estimate for the true population mean arrival time delay using confidence interval analysis.

a) Based on your random sample of 250 rows, formulate a point estimate for the mean arrival time.

b) Based on your random sample of 250 rows, generate a 95% confidence interval for the mean arrival time delay.
c) Formulate a one-sentence interpretation of this confidence interval. Imagine you are explaining the meaning of this confidence interval to someone that does not have a statistical background.

SECTION 3: Confidence Interval for Proportion (15 points total, 5 points for each part)

Imagine you are an engineer at the Department of Transportation, and you are tasked to derive an estimate for the true population proportion of late arrivals using confidence interval analysis. For this analysis, we will assume a flight is late if the arrival time delay **is greater than 10 minutes.**

a) Based on your random sample of 250 rows, formulate a point estimate for the proportion of late arrivals.
b) Based on your random sample of 250 rows, generate a 95% confidence interval for the proportion of late arrivals.
c) Formulate a one-sentence interpretation of this confidence interval. Imagine you are explaining the meaning of this confidence interval to someone that does not have a statistical background.

SECTION 4: Hypothesis Test for Mean Based on One Sample (20 points total, 4 points for each part)

Imagine you are an engineer at the Department of Transportation, and a manager makes the following claim "The average flight arrival delay time for all U.S. airlines is less than 8 minutes." You are interested to evaluate this claim using a hypothesis test method. Assume a significance level of 0.05.

a) Set up the null and alternative hypothesis. Be sure to set this up using H0 and H1 notation and use correct symbols for the population parameter of interest.
b) Calculate the test statistic.
c) Calculate the corresponding p-value.
d) Clearly state your decision as to whether you "reject the null hypothesis" or "do not reject the null hypothesis." Explain the rationale for your decision.
e) Compose an email to the manager explaining your findings. Describe your test methodology and formulate a concluding statement regarding the manager's claim. Do you support the manager's claim, or do you not support the claim? Explain the rationale for your decision in language that the manager can understand and assume the manager does not have a statistical background.

SECTION 5: Hypothesis Test for Proportion Based on One Sample (20 points total, 4 points for each part)

Imagine you are an engineer at the Department of Transportation and a manager makes the following claim "The proportion of late flights for all U.S. airlines is less than 28%." You are interested to evaluate this claim using a hypothesis test method. Assume a significance level of 0.05. For this analysis, we will assume a flight is late if the arrival time delay **is greater than 10 minutes**.

d) Set up the null and alternative hypothesis. Be sure to set this up using H0 and H1 notation and use correct symbols for the population parameter of interest.

e) Calculate the test statistic.
f) Calculate the corresponding p-value.
g) Clearly state your decision as to whether you “reject the null hypothesis” or “do not reject the null hypothesis.” Explain the rationale for your decision.
h) Compose an email to the manager explaining your findings. Describe your test methodology and formulate a concluding statement regarding the manager’s claim. Do you support the manager’s claim, or do you not support the claim? Explain the rationale for your decision in language that the manager can understand and assume the manager does not have a statistical background.

SECTION 6: Hypothesis Test for Mean Based on Two Samples (30 points total, 6 points for each part)

Imagine you are an engineer at a certain airline, and you want to investigate whether the mean arrival time delay for your airline is better than the mean arrival delay time for a competitor airline. You are interested to evaluate this claim using a hypothesis test method. Assume a significance level of 0.05.

You will select two airlines at random for this analysis. Please review the Bureau of Transportation Statistics list of airline codes (USDOT) (Access the link on the Canvas page.).

Use the first letter of your ***first name*** to select the first airline. Select the name of the airline that begins with the letter corresponding to the first letter of your first name. If there is no matching airline name corresponding to the first letter of your first name, then pick the name of the airline that most closely matches. For example, if your first name is James, you can select “JetBlue Airlines.” If your first name is Charles, you can select “Delta Airlines.” Consider this to be the airline where you are employed.

Use the first letter of your ***last name*** to select the second airline. Select the name of the airline that begins with the letter corresponding to the first letter of your last name. If there is no matching airline name corresponding to the first letter of your last name, then pick the name of the airline that most closely matches. For example, if your last name is Anderson, you can select “American Airlines.” If your last name is Rogers, you can select “Southwest Airlines.” Consider this to be the competitor airline.

i) Set up the null and alternative hypothesis. Be sure to set this up using H0 and H1 notation and use correct symbols for the population parameter of interest.
j) Calculate the test statistic.
k) Calculate the corresponding p-value.
l) Clearly state your decision as to whether you “reject the null hypothesis” or “do not reject the null hypothesis.” Explain the rationale for your decision.
m) Compose an email to the CEO of your airline explaining your findings. Describe your test methodology and formulate a concluding statement regarding whether the mean arrival time delay for your airline is better than the mean arrival delay time for a competitor airline. Explain the rationale for your decision in language that the CEO can understand and assume the CEO does not have a statistical background.

REMINDER:

- Enter your answers in the Word Template file and submit your Excel file showing all calculations and graphs.
- For supplemental assistance, please review the video resources provided in Module 6 (confidence intervals), Module 7 (hypothesis testing for one sample), and Module 8 (hypothesis testing for two samples).
- Refer to the grading rubric provided in Canvas for this assignment.